单缝夫琅禾费衍射

“光学”模块的知识结构

● 细化的“课程知识点”

- 几何光学基本定律（教材11.14节）、光程、费马原理、薄透镜成像（1次课）
- 相干光、光程差、杨氏双缝干涉；菲涅尔双镜*、劳埃德镜*（1次课）
- 薄膜干涉、等倾干涉；等厚干涉：劈尖干涉、牛顿环、迈克尔孙干涉仪*（2次课）
- 惠更斯-菲涅尔原理，单缝夫琅禾费衍射、半波带法；圆孔衍射、光学仪器的分辨本领(2次课)
- 衍射光栅、光栅方程、光栅衍射的光强分布、缺级现象(1次课)
- 光的偏振性、偏振片、起偏和检偏、马吕斯定律、反射折射起偏、布儒斯特定律(1次课)
- 晶体双折射现象、惠更斯作图法、偏振棱镜、1/4波片和半波片、偏振光干涉(1次课)

● 主要拓展内容

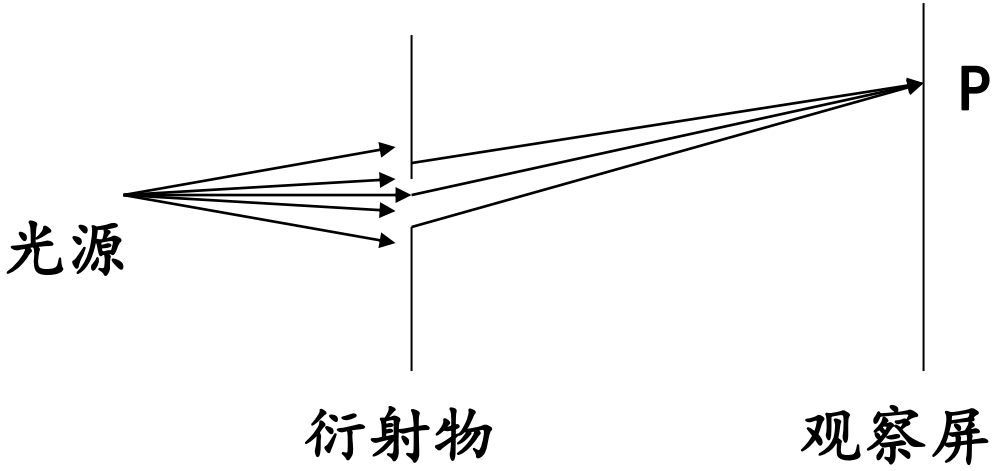
拓展：大口径光学望远镜及相关技术介绍

拓展：X射线衍射及应用（大物D）

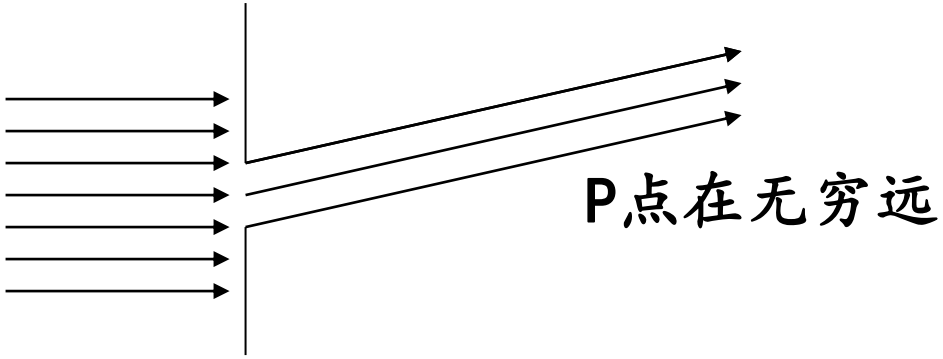
拓展：晶体双折射现象、惠更斯作图法、偏振棱镜、1/4波片和半波片、偏振光干涉

1.衍射现象 光波遇到障碍物时，绕过障碍物进入几何阴影区。

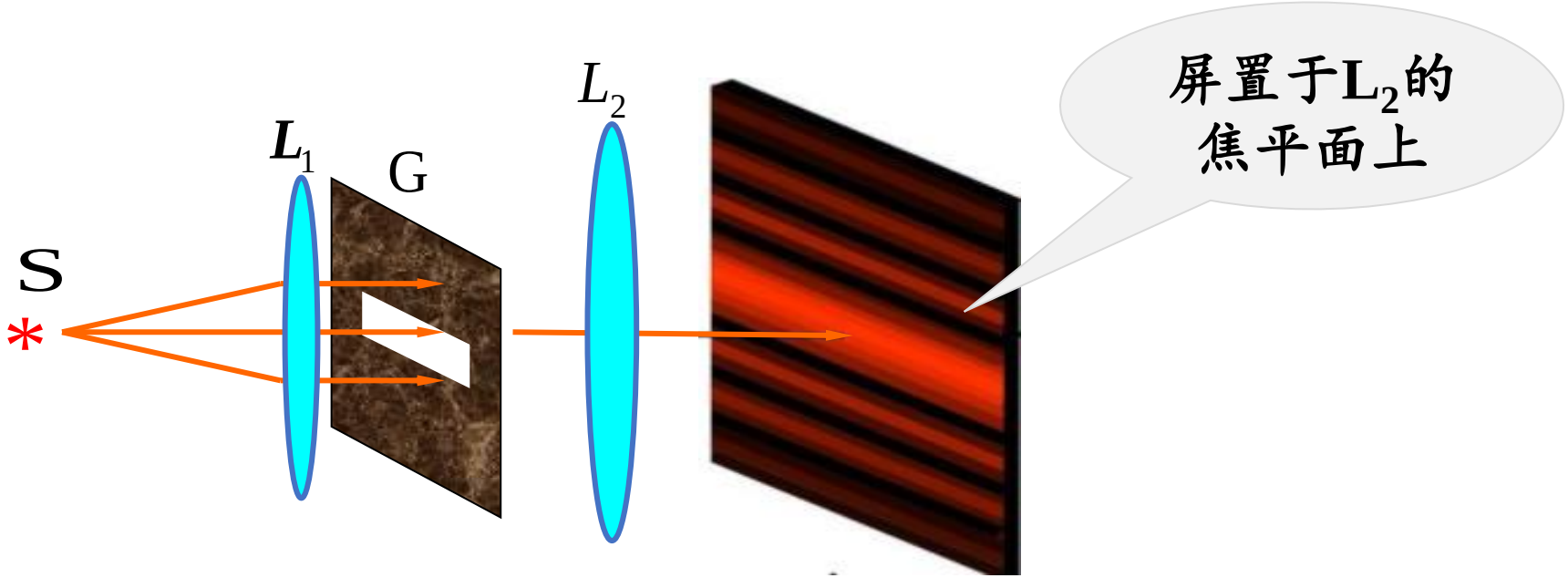
2.菲涅耳衍射



3.夫琅禾费衍射

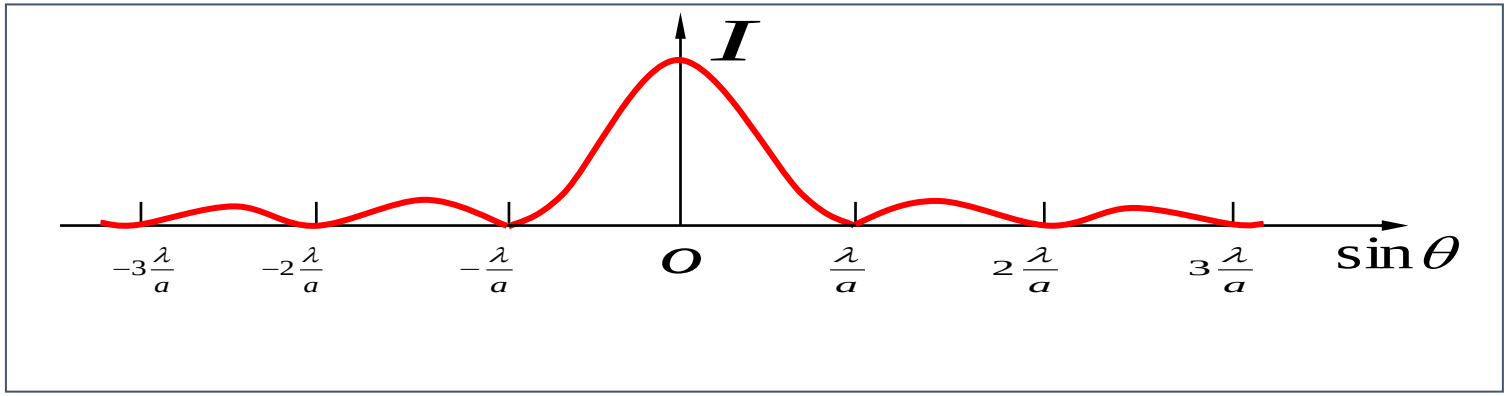


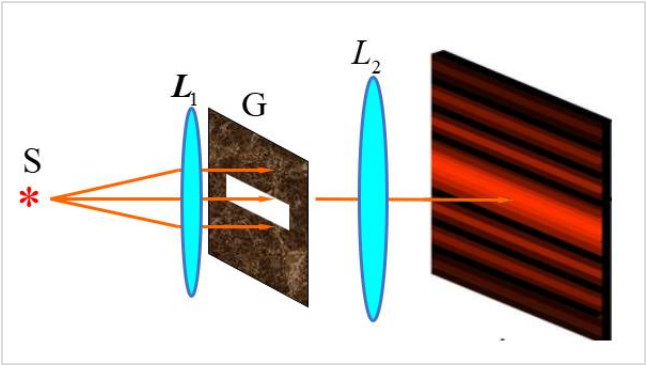
1. 实验装置



2. 实验现象

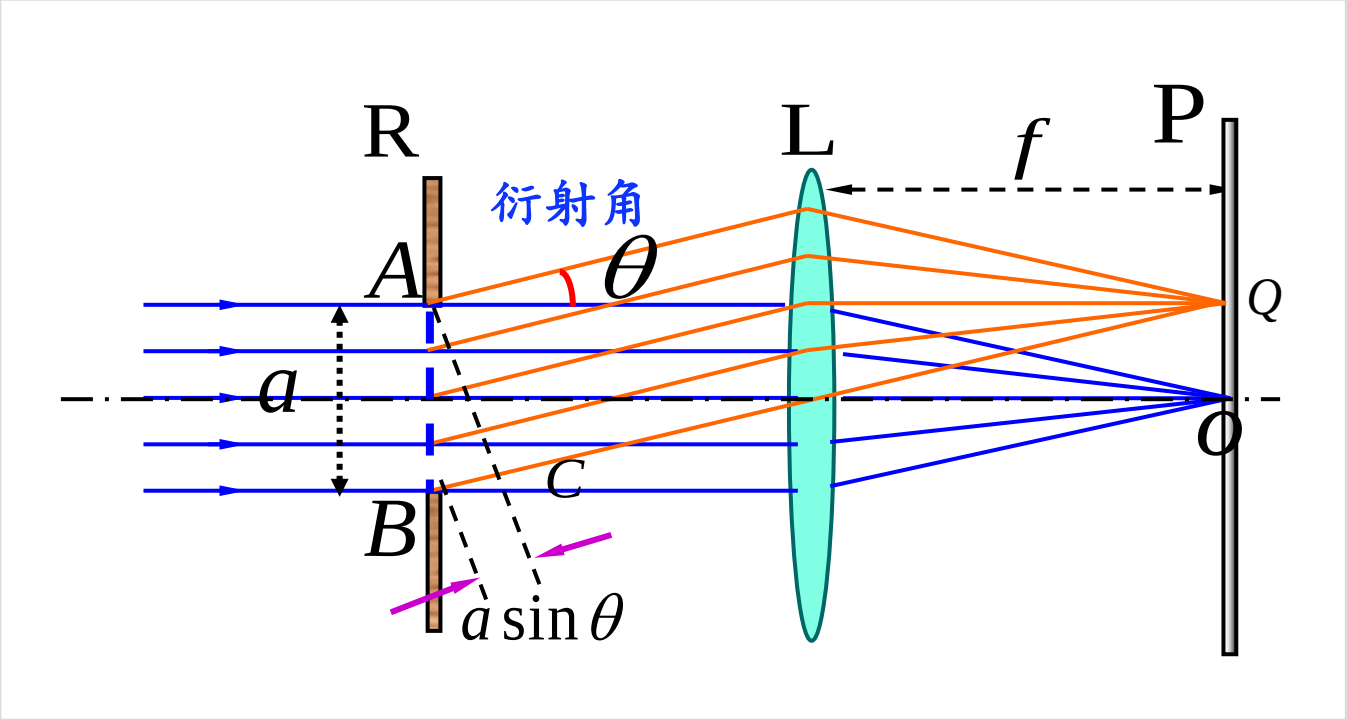
明暗相间的平行直条纹，条纹的宽度和亮度不同





θ 为衍射角

1. 物理图像



- 衍射角θ（向上为正，向下为负）
- 狭缝两条边缘衍射线之间的光程差 $BC = a\sin\theta$
- Q 点条纹的明暗取决于光程差 BC 的大小

2. 菲涅尔半波带

$BC = a\sin\theta$

$$a\sin\theta = 0$$

—— 中央明纹

$$a\sin\theta = \pm 2k\frac{\lambda}{2}$$

$2k$个半波带

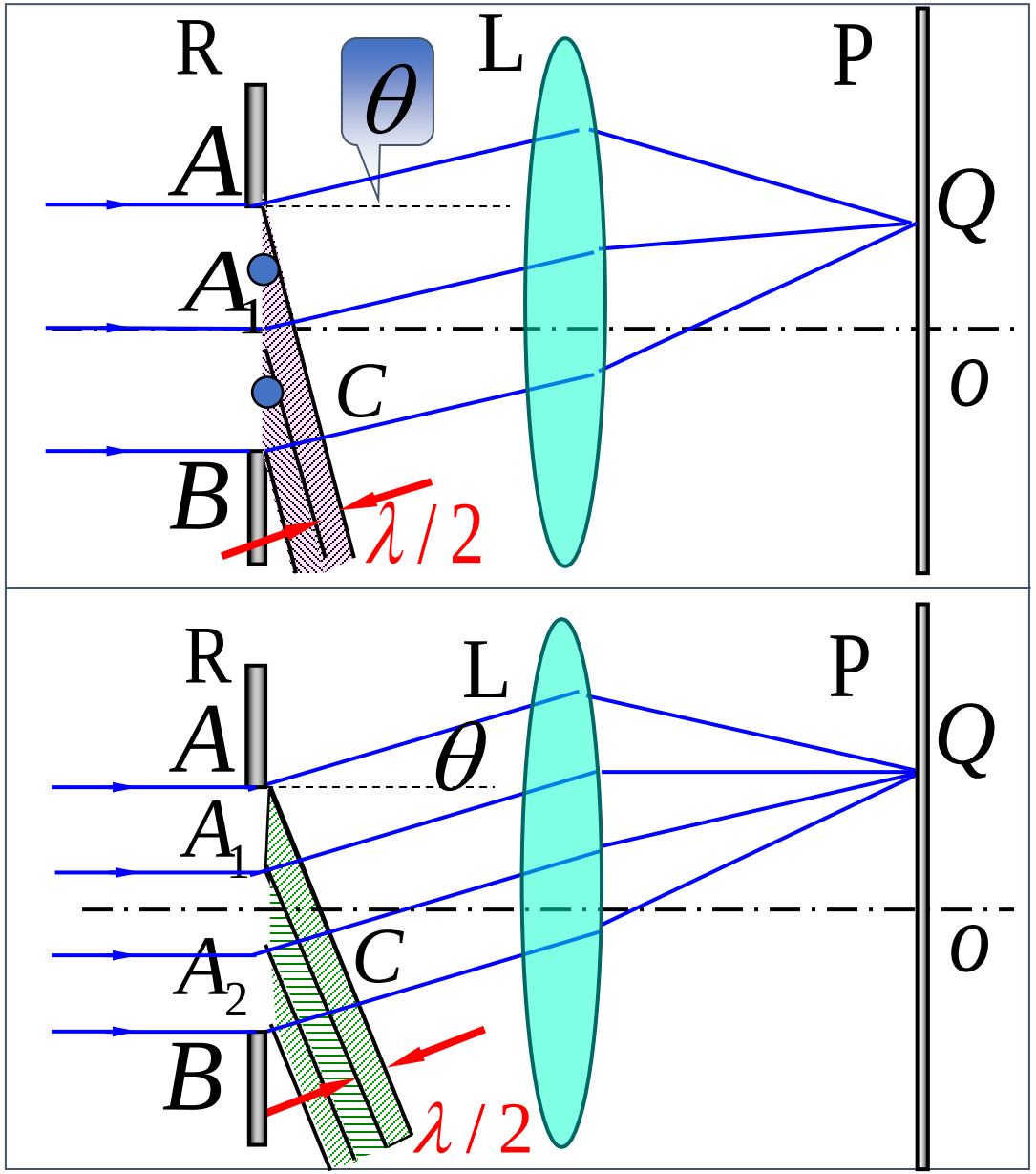
—— 暗条纹

$$a\sin\theta = \pm(2k+1)\frac{\lambda}{2}$$

$2k+1$个半波带

—— 明条纹

$$k = 1, 2, 3, \cdots$$

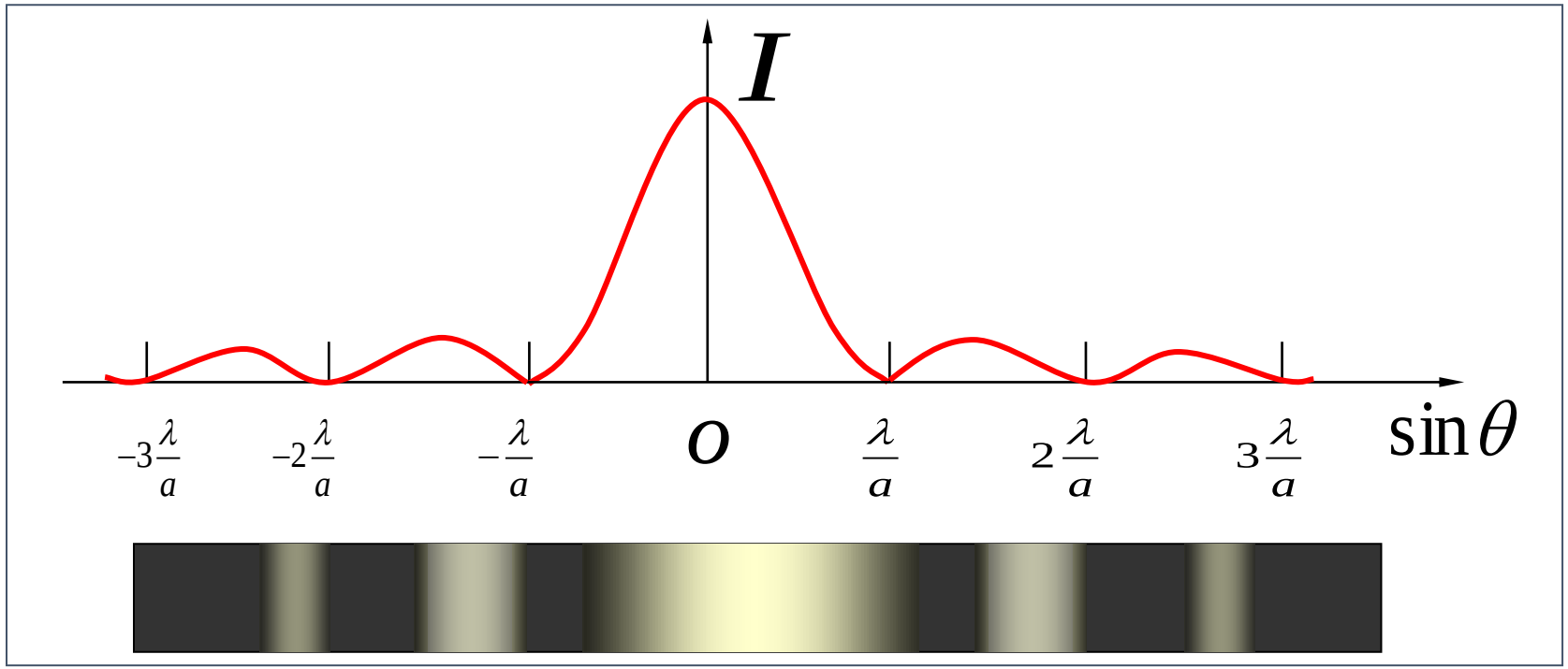


3. 光强分布

$$a\sin\theta = \pm 2k\frac{\lambda}{2}.$$ **干涉相消（暗纹）**

$$k = 1, 2, 3, \cdots$$

$$a\sin\theta = \pm(2k+1)\frac{\lambda}{2}.$$ **干涉加强（明纹）**

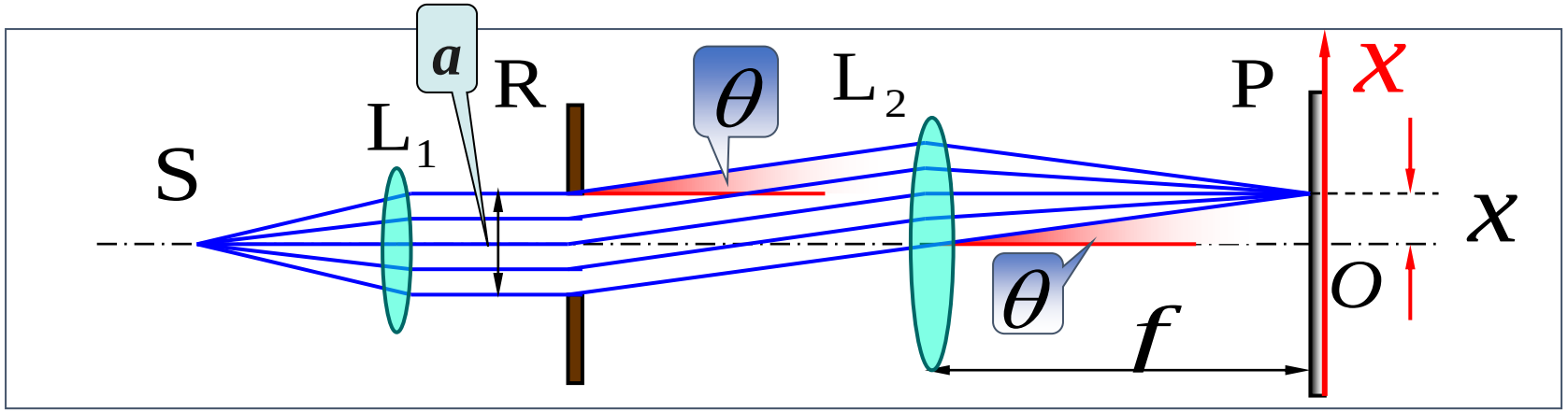


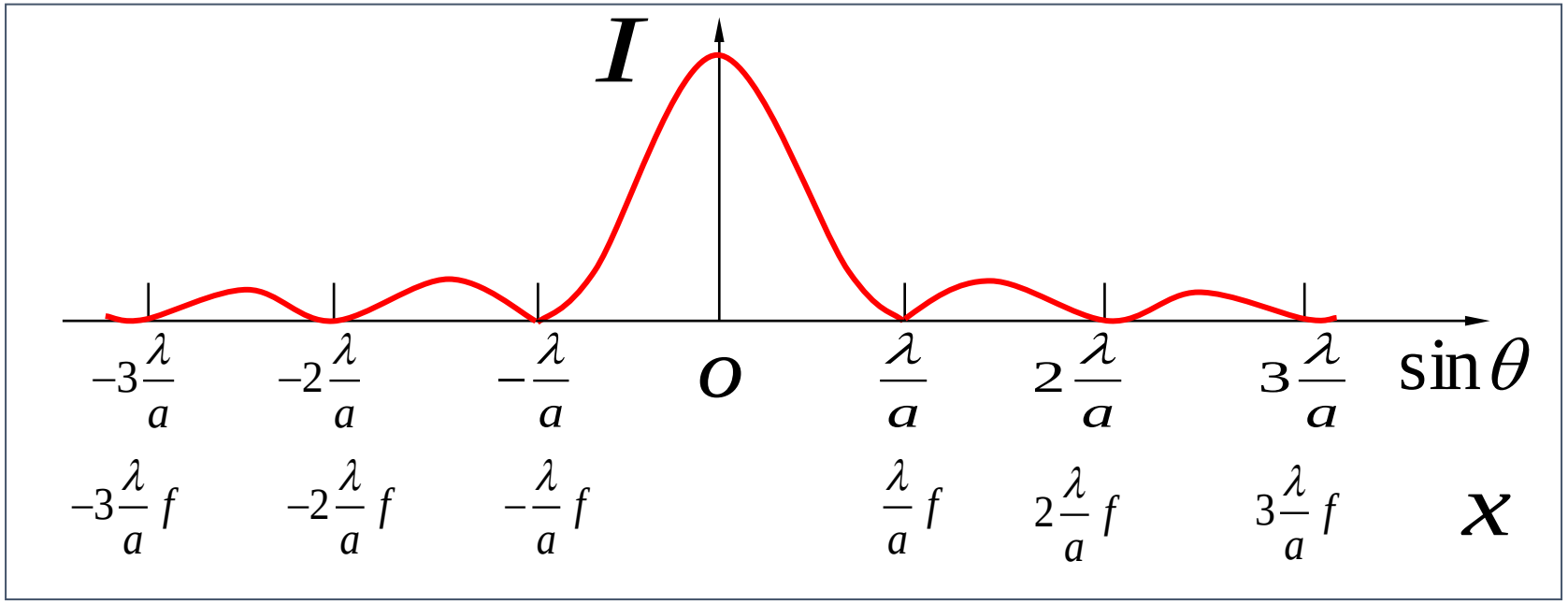
3. 光强分布

当 θ 较小时，

$$\sin\theta \approx \theta$$

$$x = \theta f$$





1. 第一暗纹的衍射角

$$\theta_1 = \arcsin\frac{\lambda}{a}.$$

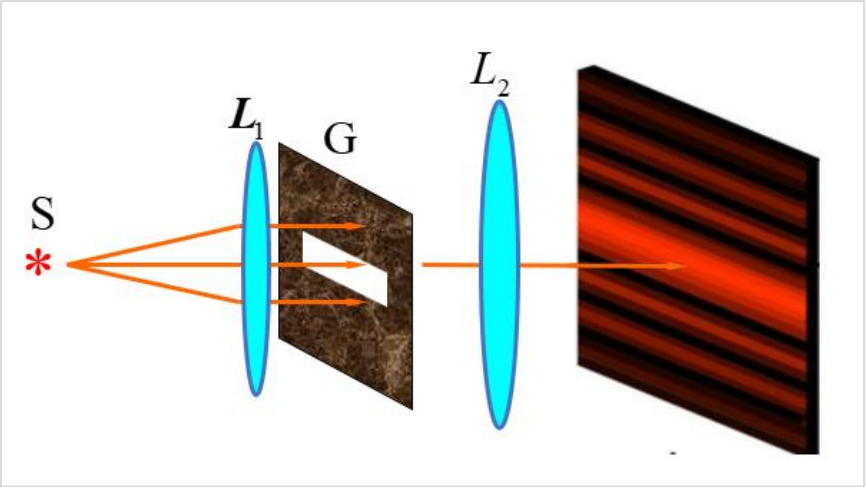
(1) 波长一定时，a增大 → θ_1减小

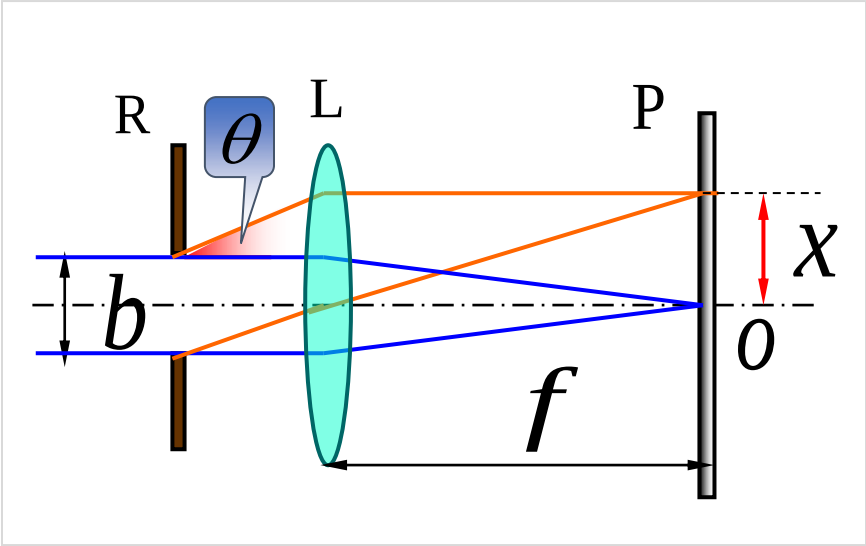
$$\frac{\lambda}{a} \to 0,\ \theta_1 \to 0.$$ （直线传播）

$$\frac{\lambda}{a} \to 1,\ \theta_1 \to \frac{\pi}{2}.$$ （衍射最强）

(2) 狭缝宽度一定时，λ 越大 → θ_1 也变大

（波长越长，衍射越明显）





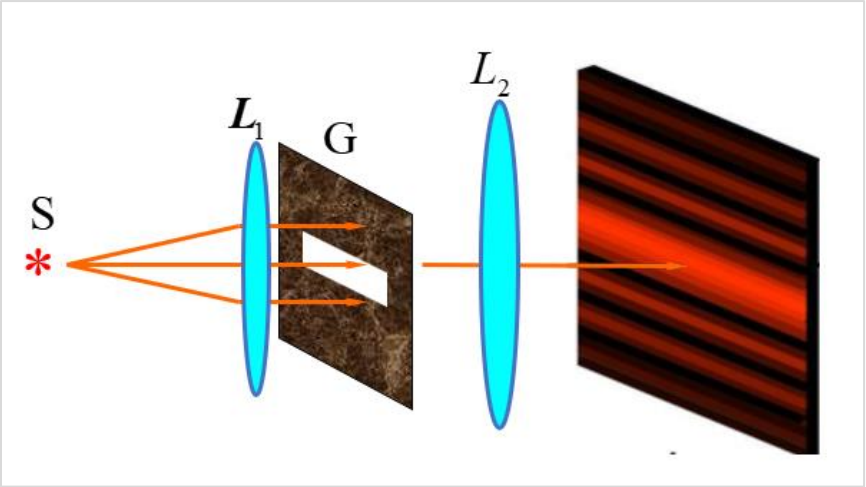
2. 中央明纹 （$k=1$ 的两个暗纹之间）

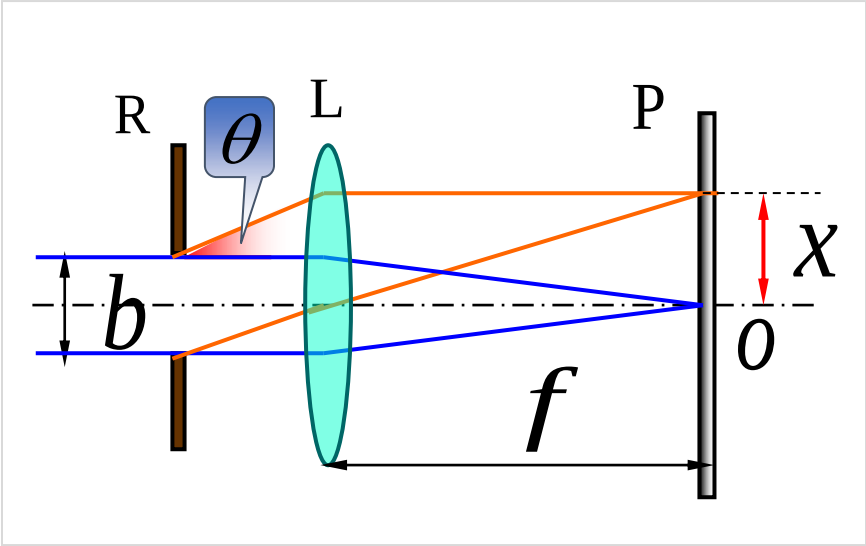
角范围 $$-\frac{\lambda}{a}<\sin\theta<\frac{\lambda}{a}$$

线范围 $$-\frac{\lambda}{a}f<x<\frac{\lambda}{a}f$$

中央明纹的宽度：

$$l_0=2x_1\approx 2\frac{\lambda}{a}f.$$





3. 条纹宽度 （相邻条纹间距）

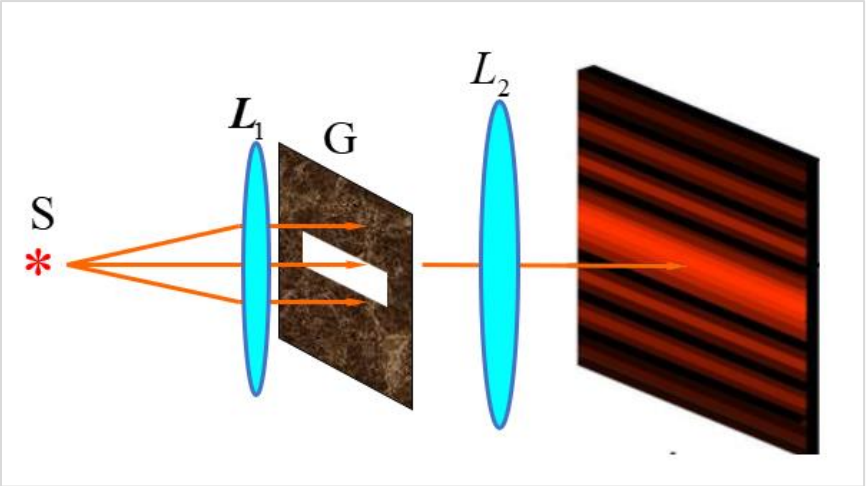
$$a\sin\theta = \pm 2k\frac{\lambda}{2}.$$ 干涉相消（暗纹）

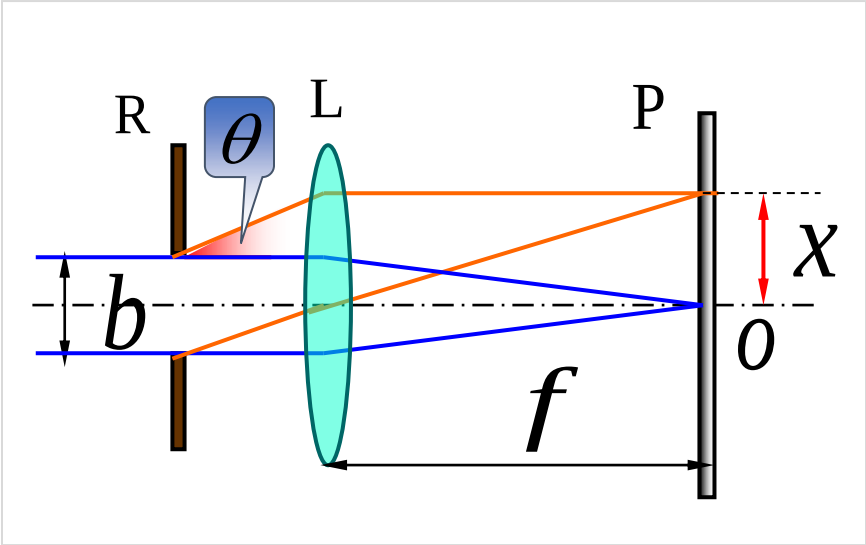
$$a\sin\theta = \pm(2k+1)\frac{\lambda}{2}.$$ 干涉加强（明纹）

$$k = 1, 2, 3, \cdots$$

其他明纹宽度(中央明纹除外)：

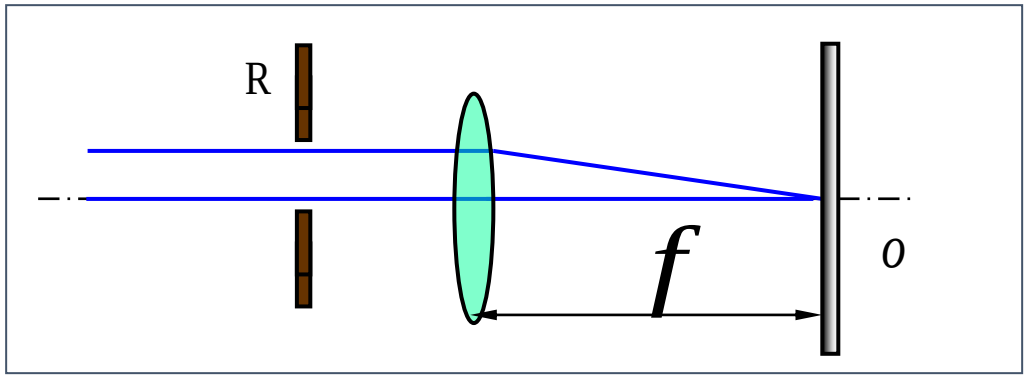
$$l \approx \frac{l_0}{2} \approx \frac{\lambda}{a} f.$$





4. 条纹动态变化分析

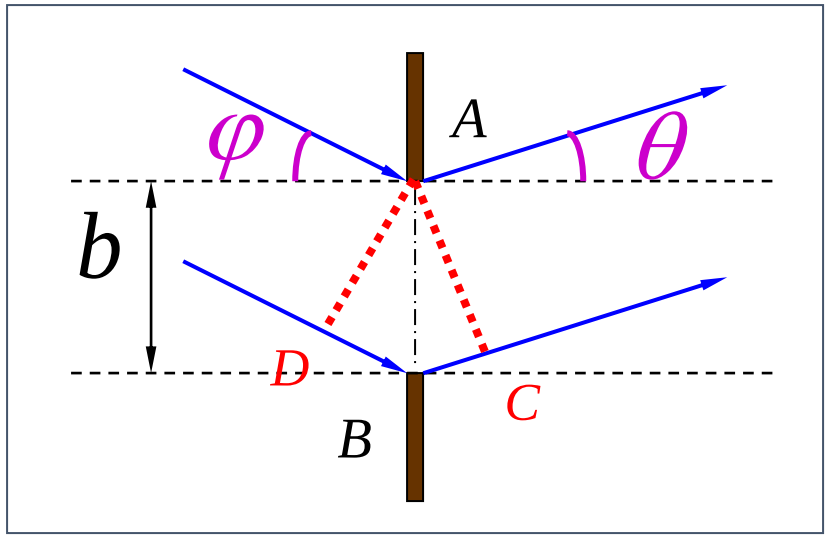
- 若狭缝上下微小移动



衍射条纹位置不发生变化。

5. 入射光非垂直入射时光程的计算

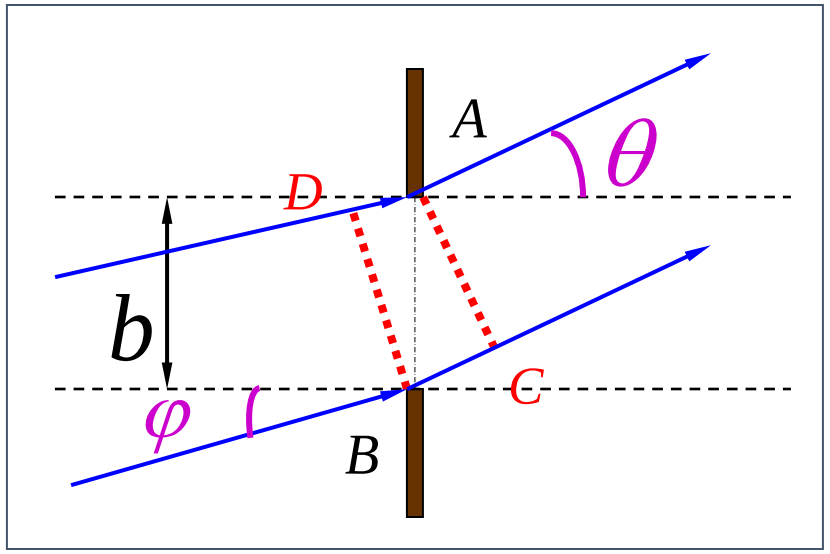
（1）如下图时：



$$\Delta = DB + BC = b(\sin\theta + \sin\varphi)$$

（中央明纹向下移动）

（2）如下图时：



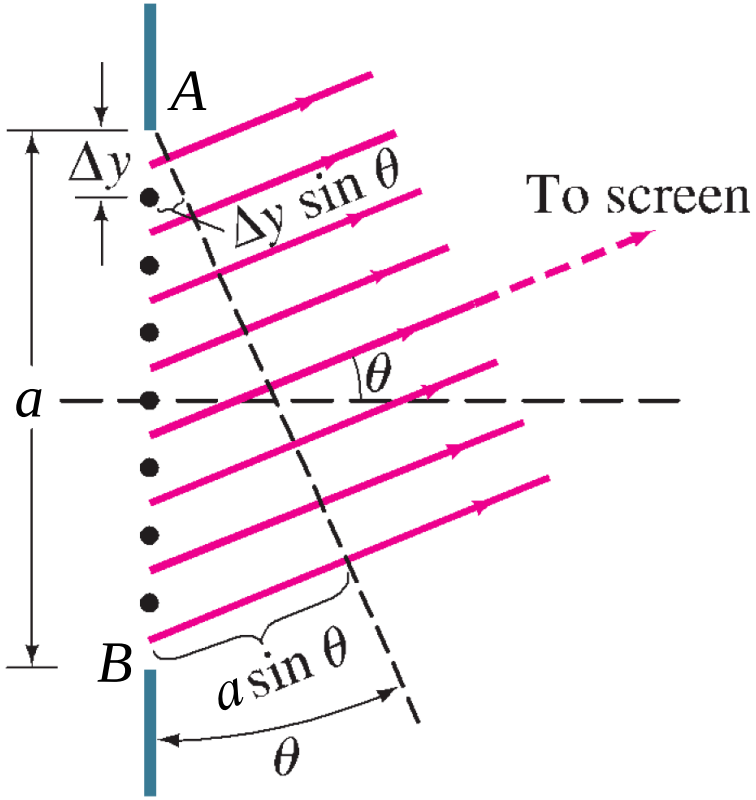
$$\Delta = BC - DA = b(\sin\theta - \sin\varphi)$$

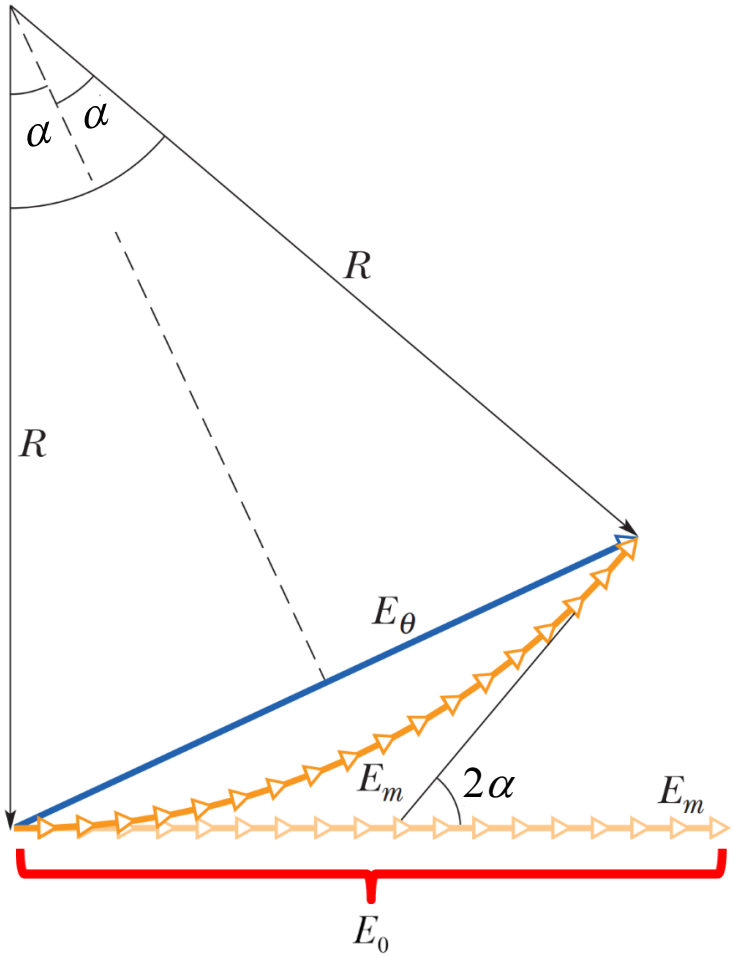
（中央明纹向上移动）

6. 矢量图解分析法

将单缝 a 分成N个宽度为 Δy 的条带（$N \to \infty$）：

- 相邻条带光程差 $\Delta y \sin\theta$.
- 光线 B 相对于 A 总的光程差为 $a\sin\theta$.
- 总相位差：$\alpha = \dfrac{2\pi}{\lambda} a \sin\theta$.





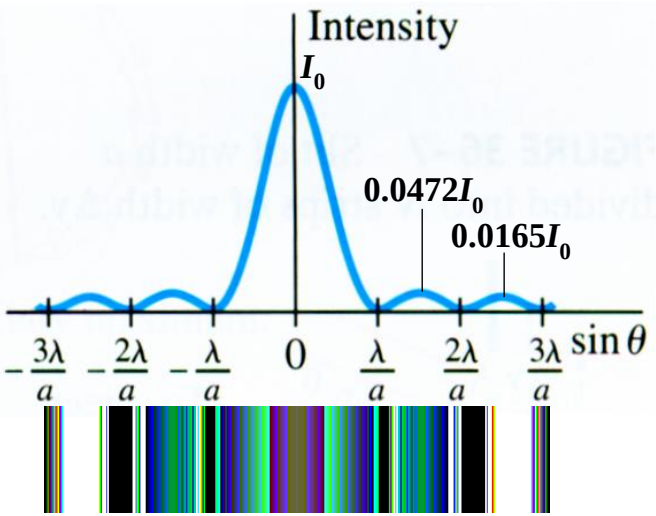
单缝夫琅禾费衍射的矢量图解分析法（旋转矢量法）

6. 矢量图解分析法

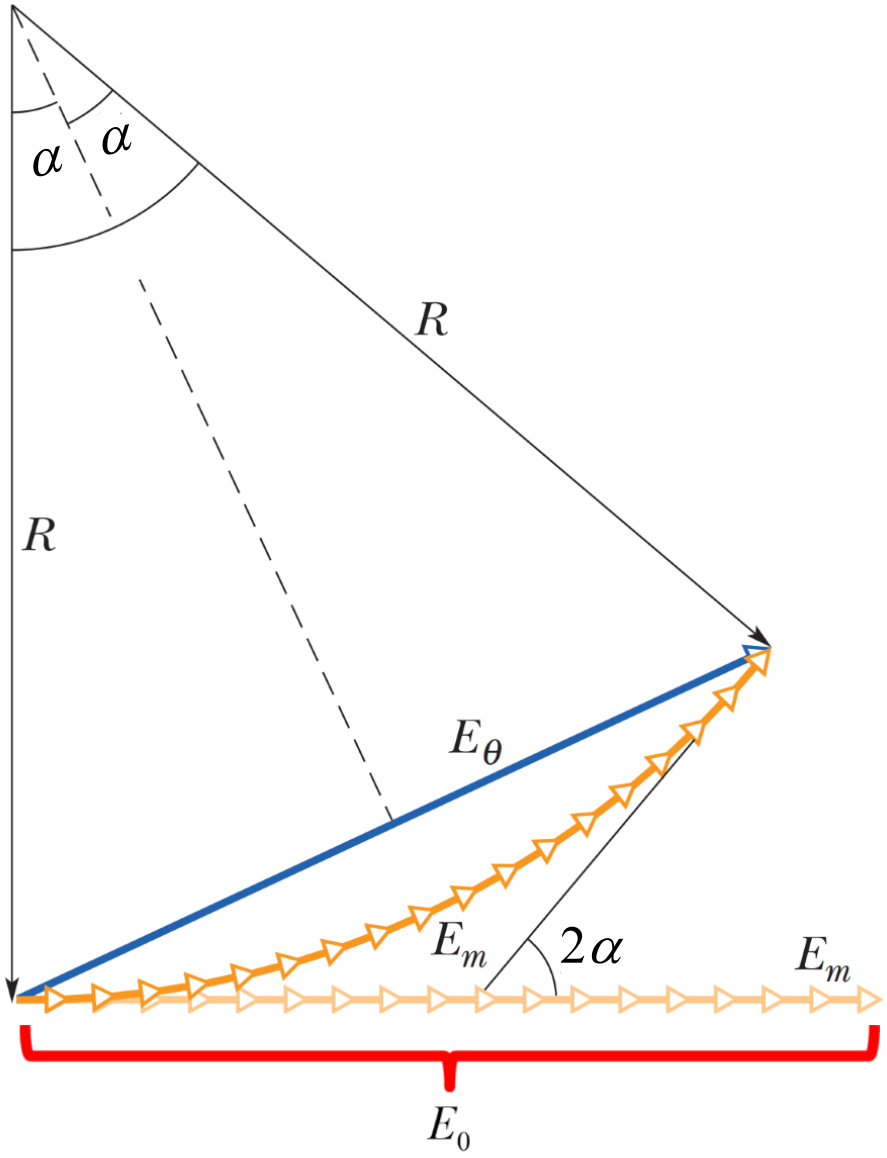
根据几何关系： $E_0 = \lim_{N\to\infty}\sum_m E_m = 2R\alpha;$

$$E_\theta = 2R\sin\alpha$$

简单推导可得： $E_\theta = E_0\dfrac{\sin\alpha}{\alpha}$ $\qquad I = I_0\left(\dfrac{\sin\alpha}{\alpha}\right)^2$

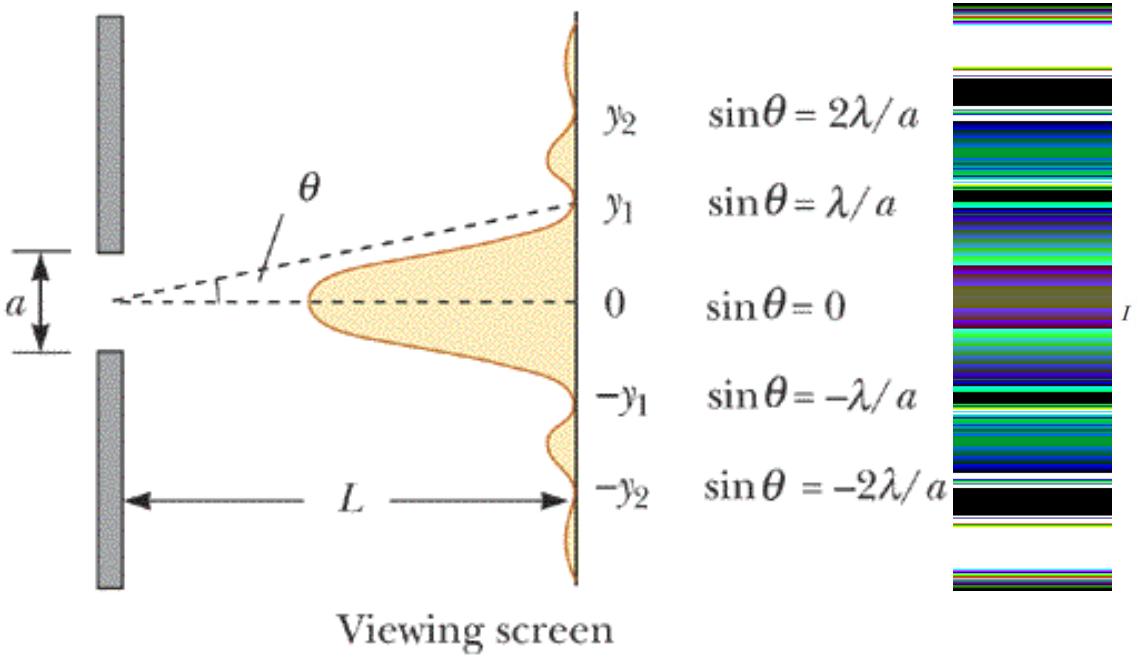


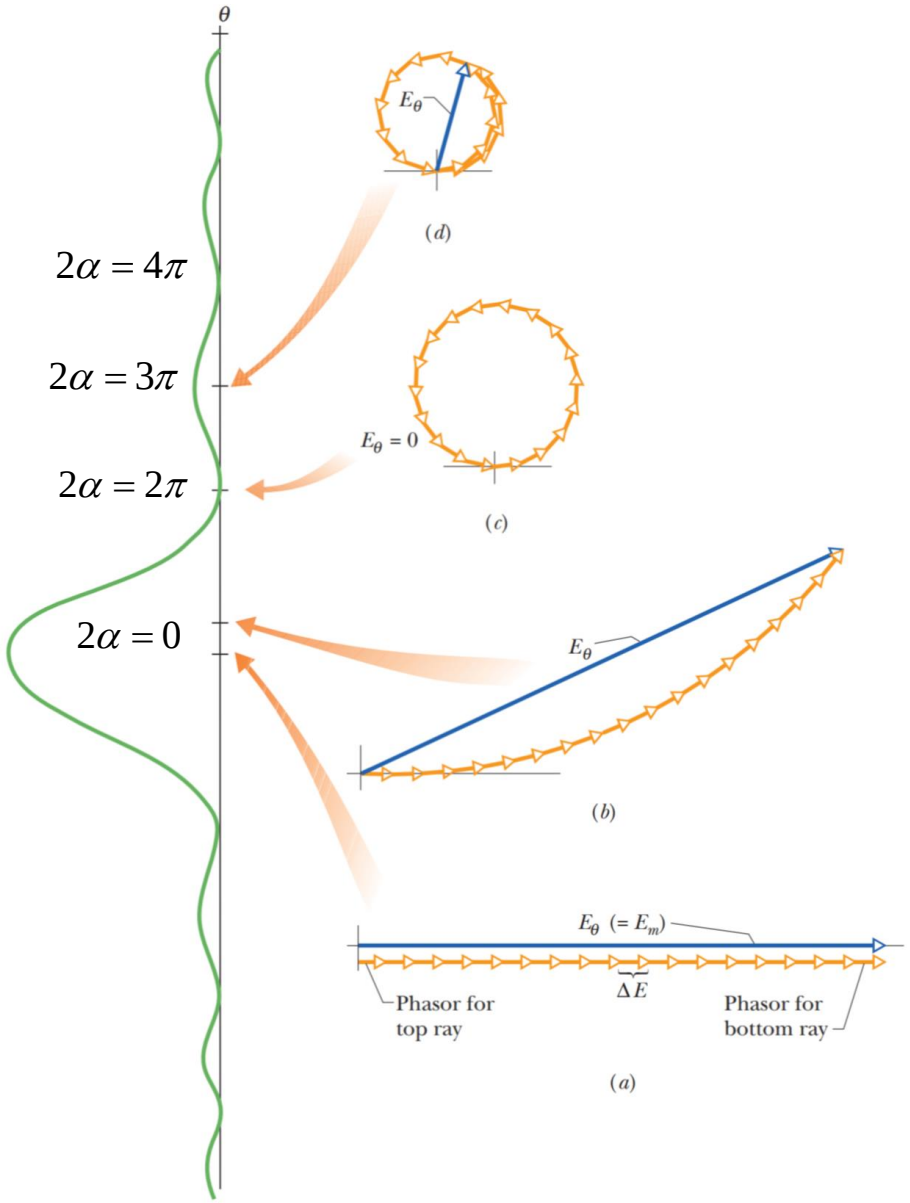
$$\alpha = \frac{2\pi}{\lambda}a\sin\theta$$



6. 矢量图解分析法

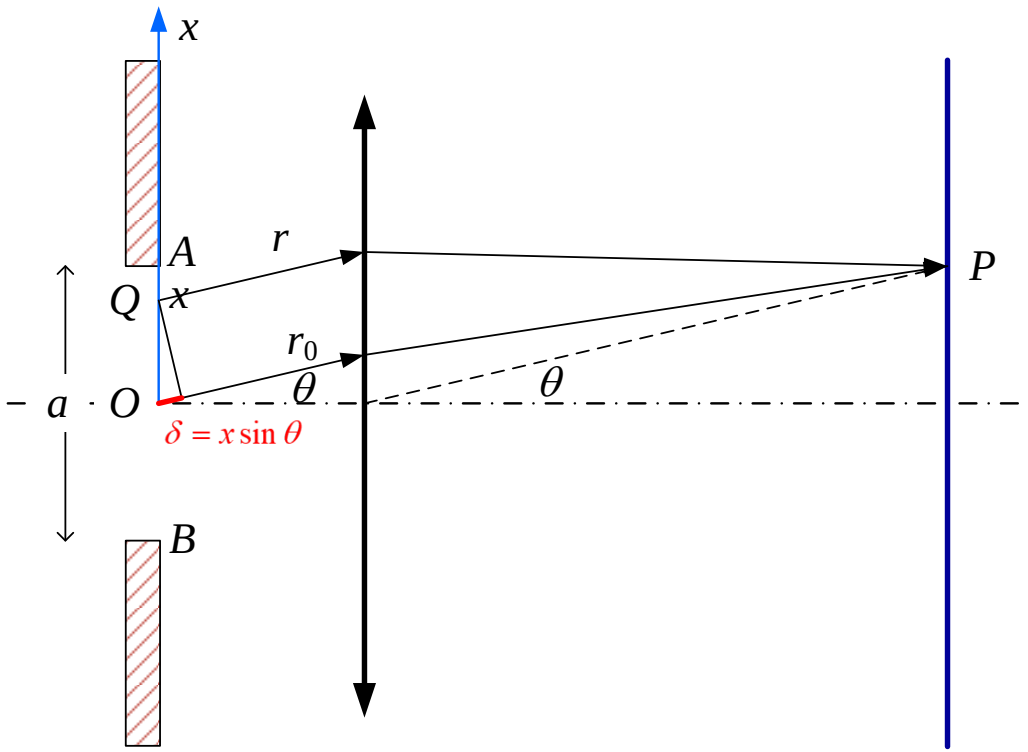
$$I = I_0\left(\frac{\sin\alpha}{\alpha}\right)^2$$





7. 惠更斯菲涅耳原理积分法 *

$$U(P)=\frac{1}{i\lambda}\iint U(Q)\frac{e^{ikr}}{r}\frac{\cos\theta_0+\cos\theta}{2}d\Sigma$$



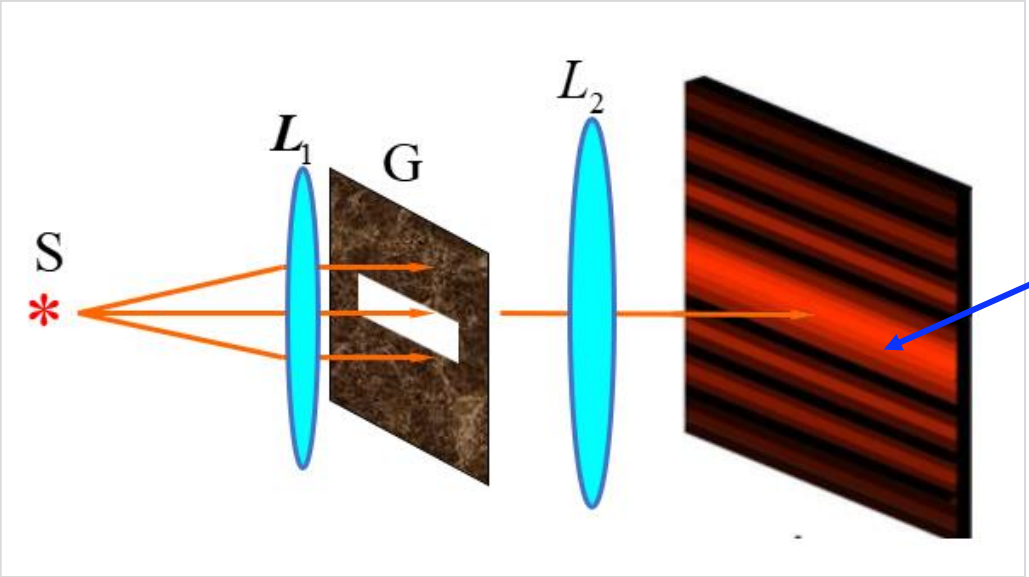
- 单缝长 b，因此： $d\Sigma = bdx$

- 取近似：分母上 r 用 r_0 替代，相位上 r 处理为 $r = r_0 - x\sin\theta$ 。

$$\rightarrow \quad U(P)=\frac{bU_0}{i\lambda r_0}\int_{-a/2}^{a/2}\exp\left[ik\left(r_0-x\sin\theta\right)\right]dx$$

$$=\frac{bU_0}{i\lambda r_0}\frac{e^{ikr_0}}{-i\sin\theta}\left[e^{-i\frac{a}{2}\sin\theta}-e^{i\frac{a}{2}\sin\theta}\right]=C\frac{\sin\left(\frac{\pi a\sin\theta}{\lambda}\right)}{\left(\frac{\pi a\sin\theta}{\lambda}\right)}=C\frac{\sin\alpha}{\alpha}.$$

得证。



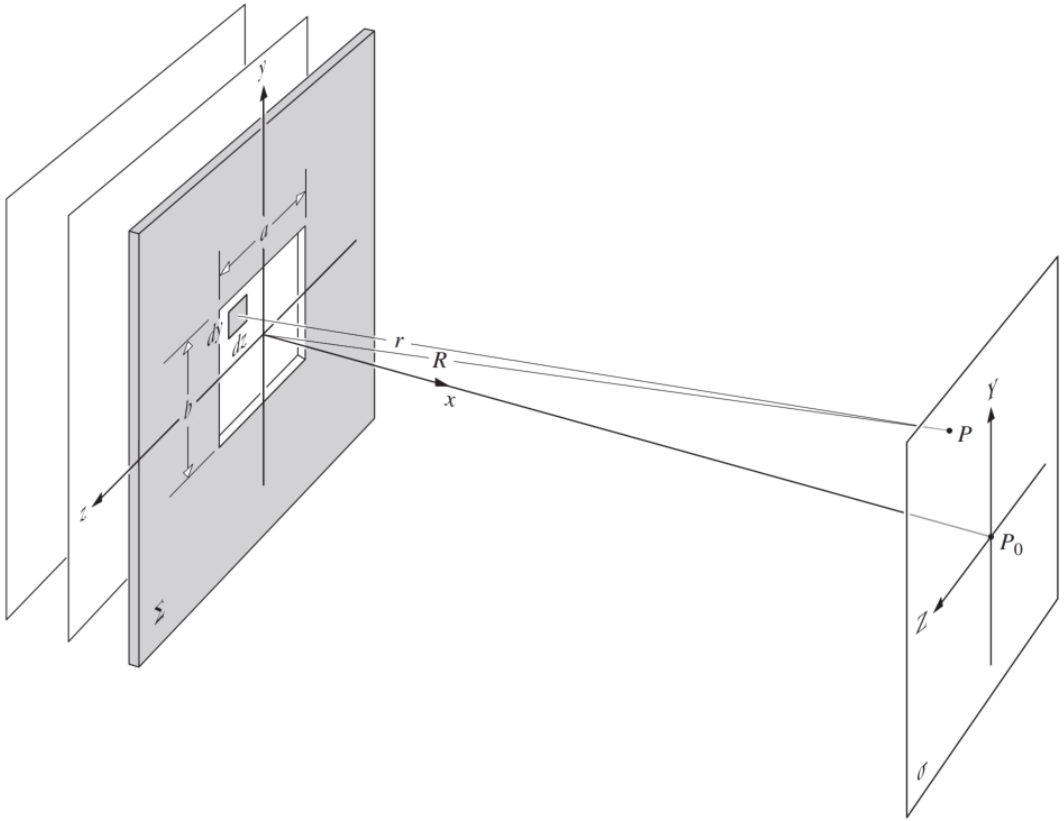
零级衍射斑内集中了80% 以上的能量

$$-\frac{\lambda}{a} < \sin\theta < \frac{\lambda}{a}$$

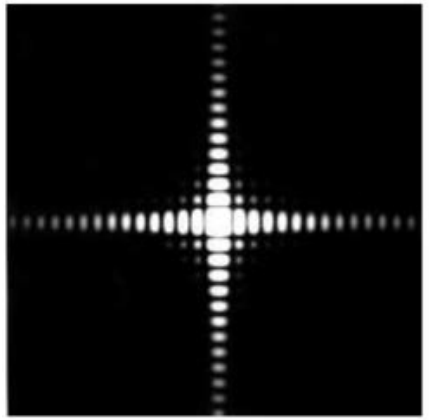
1. 几何光学是波动光学在 $\lambda/a \to 0$ 时的极限情形。

2. 缝极细（$a \approx \lambda$）时，衍射屏也看不到单缝衍射条纹。

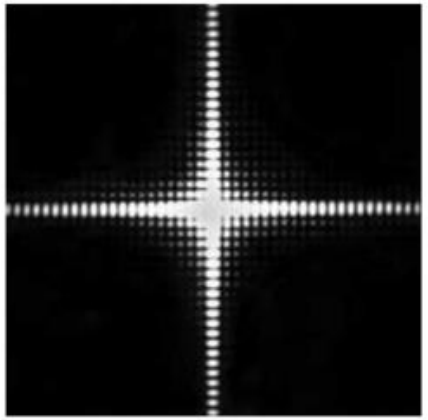
3. 干涉和衍射都是波的相干叠加，本质上没有区别。



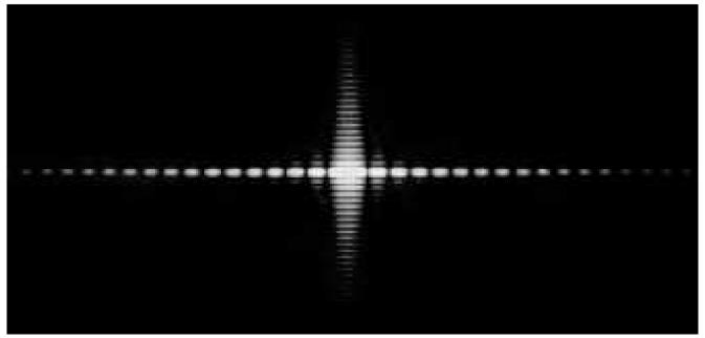
正常曝光



曝光过度



$b > a$ 情况



矩孔夫琅禾费衍射 *

→ 圆矩孔夫琅禾费衍射 ???

谢　谢

光的衍射典型例题（一）

例1. 波长为500nm的平行光垂直照射在单缝上。(1) 第一级暗纹的衍射角为30°，求单缝宽度。(2) 若单缝宽度为0.5 mm，缝后紧挨着的薄透镜焦距为1m，求：中央亮纹角宽度；中央亮纹线宽度；第一级与第二级暗纹的距离。(3) 在(2)的条件下，若在屏幕离中央亮纹中心3.5mm处 p 点为一亮纹，求：p 处亮纹的级数；从 p 处看，对于该光波，狭缝处的波阵面可分割几个半波带。

解： (1) $a\sin\theta = \pm k\lambda \quad \rightarrow \quad a = 1\mu\text{m}.$

(2) $$\Delta\theta_0 \approx \frac{2\lambda}{a} = \frac{2\times500\times10^{-9}}{0.5\times10^{-3}} = 2\times10^{-3}\text{rad}.$$

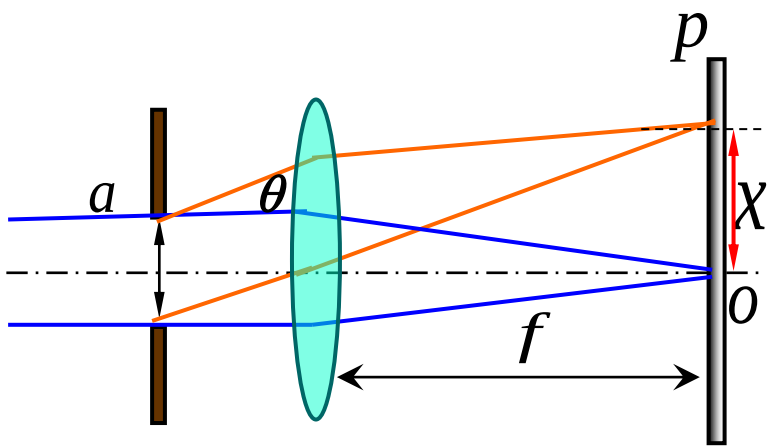
（中央亮纹角宽度）

$$\Delta x_0 \approx f\Delta\theta_0 = 2\text{mm}.$$

（中央亮纹线宽度）

$$\Delta x \approx \frac{1}{2}\Delta x_0 = 1\text{mm}.$$

（第一级与第二级暗纹的距离）



光的衍射典型例题（一）

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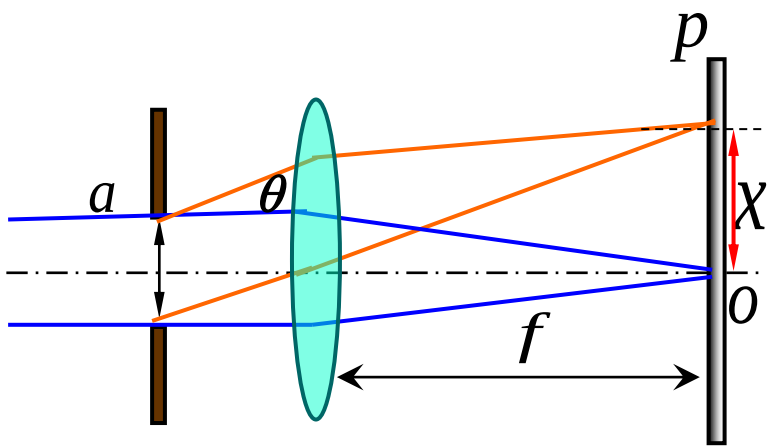
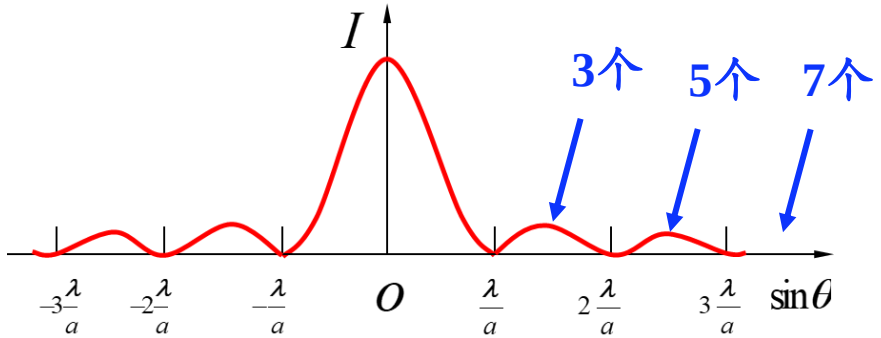
解： (3) $\sin\theta \approx tg\theta \approx \theta = \dfrac{x}{f}.$

$$a\sin\theta = (2k+1)\frac{\lambda}{2}$$ （亮纹条件，k = ?）

$$k = \frac{ax}{\lambda f} - \frac{1}{2} = \frac{0.5\times 3.5}{5\times 10^{-4}\times 1\times 10^{3}} - \frac{1}{2} = 3.$$

（k =3，对应7个半波带）